AAE 33300 Fluid Mechanics
3 Credits

1 Course Information/Description

This course is about discovering the secrets of fluid mechanics – encountered in everyday life as well as in every significant human technology – including the secrets of flight. The main goal of this course is to build the framework which will allow us to solve simple engineering problems involving fluid mechanics and give the basic understanding needed for further study in this area or to work on interdisciplinary problems involving fluid mechanics.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's Brightspace learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:

John D. Anderson, "Fundamentals of Aerodynamics," McGraw Hill (5th, 6th or 7th ed.)

3 Assignments

This course is composed of only individual assignments. The two main types of assignments are:

3.1 Weekly Homework Sets

Each week, a problem set will be assigned based on the lectures from the previous week. These assignments will challenge students to apply engineering concepts to practical problem.

3.2 Weekly Quizzes in Brightspace

Each week, a quiz based on the reading assignment for the week will be available through Brightspace.

4 Course Learning Objectives

1. Calculate hydrostatic pressure distributions, forces, and moments.
2. Calculate aerodynamic forces and moments from pressure and shear stress distributions.
3. Apply similitude to scale data.
4. Apply global conservation of mass and momentum control volume analysis to engineering systems.
5. Apply Bernoulli's equation to relate velocity and pressure.
6. Calculate velocity and pressure fields using potential theory.

7. Apply the Navier-Stokes Equations to obtain exact solutions and compute shear stress and other flow quantities.
8. Apply the Blasius Boundary Layer Solution and integral methods to obtain boundary-layer flow properties.

5 Evaluation

All assignments must be submitted via Gradescope (Homework Sets) or completed via Brightspace (Quizzes). Late homework will be accepted (unless stated otherwise) up to 11:59 pm Eastern US time on the Monday following the due date for 50% credit. No homework will be accepted later than that, and all problems must be handed in at the same time.

Your final grade will be computed two ways. First your homework, Midterm 1, Midterm 2, and the Final Exam will each be weighted at 25% of your final grade. In the second calculation, the lower of Midterm 1 and Midterm 2 will be weighted at 10%. The higher of Midterm 1 and Midterm 2 and your homework will be weighted at 25% of your grade, and your Final Exam will be counted as 40% of your final grade. You will receive the higher of these two final grades.

	Method 1	Method 2
Homework & Quizzes	25%	25%
Midterm 1	25%	25% or 10%
Midterm 2	25%	10% or 25%
Final Exam	25%	40%